PROJECT I

RELATIONSHIP BETWEEN THE CANADA–US EXCHANGE RATE AND COMMODITY PRICES

I

RELATIONSHIP BETWEEN THE CANADA–US EXCHANGE RATE AND COMMODITY PRICES

Exchange Rate Dynamics · Commodity Prices · OLS Regression

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Introduction

Owing to globalization, especially regarding trade, the relationship that exists between exchange rates and commodity prices has attracted attention in the literature. As this helps forecast better if not reduce the occurrence of loss, it will be avoided. Ferraro and Rossi (2015) from their findings on the relationship between exchange rate and commodity prices found that having known future prices of commodities, the exchange rate can be forecasted. Since their paper focused on oil prices, they arrived at a conclusion that with a good model of forecasting oil prices, it is possible to forecast the exchange rates in the future. Also, Chen et al., (2008) from their regression between the two variables found that the exchange rates were able to predict commodity prices strongly, however, the opposite was weak. Which is, the commodity prices were weak with forecasting the exchange rates. This is supported by Kohlscheen et al., (2017) who found also that the link between commodity prices and exchange rates is statistically significant at a high frequency. Also, Bashar and Kabir (2013) using cointegration and error correction models from using data from the third quarter of 1982 to the second quarter of 2013 found that the coefficients of interest rate and commodity prices in Australia were positive and significant on the exchange rate of the country in the long run. Similarly, for the Canada-US exchange rate and commodity prices, Choudhri and Schembri (2014) through empirical analysis found that there is a positive and significant long-run effect of productivity and commodity prices indices on the exchange rate, meanwhile, this has shifted and has a stronger and positive trend since 1990. Furthermore, from the empirical results of Jung et al., (2020), there were both short-run and long-run asymmetric relationships existing between the Canada-US exchange rate and the real price of oil. In addition, Mow (2013) saw that the commodity prices explain the variation in the US-Canadian dollar exchange rate as well as other currencies. These emphasize support the relationship between commodity prices and the exchange rate as is the focus of this paper especially with the Canada-US exchange rates and commodity prices.

Theoretical Framework

Put forward by Milijkovic (1999), the exchange rate is referred to in simple language as the local or home-currency price of a foreign currency, that is quoting the foreign currency in the face of the home currency. And the commodity prices are the set prices for commodities and items as determined by the forces of demand and supply in the market. In consequence, the analysis of the relationship between these two variables stems from the "law of one price". According to Isard, P. (1977), the law of one price prevails when there are no transport costs, trade barriers, the perfect commodity arbitrage give way for uniform pricing of goods in common currency units throughout the world. More so, "the Law of one price states that once prices are converted to a common currency, the same good should sell for the same price in different countries", Milijkovic (1999, pg.126). Therefore, if a particular good is valued in a currency for trade common to different economies, the price or value should not differ within these different economies with the same currency, but they all should have a uniform price – "one price" across. Notwithstanding, there are arguments in the literature that the "law of one price is violated, [the likes of, Richardson, 1978; Isard 1977]. However, Rogoff found that the Law of one price applies better with countries that engage with highly traded commodities, for example, gold. Therefore, this law is referred to as an essential ingredient in the field of international economics and trade whose absence implies no full theory of international trade, as opined by Officer (1986). Meanwhile, the Canadian dollar which is the focus of this paper, according to Laidler and Aba (2001) appreciated against some currencies and depreciated against others from the findings of their paper from 1973-2000. From the paper, it was concluded the Canadian dollar is a commodity currency as its value depends on the changes in commodity prices. Hence, this paper explores the Canadian dollar, a commodity currency, and its relationship with the US dollar in commodities and exchange rates.

Description of Data

The data for this project comes from Statistics Canada. The commodity price indices are in Table 10-10-0132-01 and exchange rates are in Table 10-10-0009-01. Use MONTHLY frequency, where the prices were determined using the Fisher commodity price index, the exchange rate reflects Canadian cents per US dollar, and the observations were made monthly from January 1972 to April 2017.

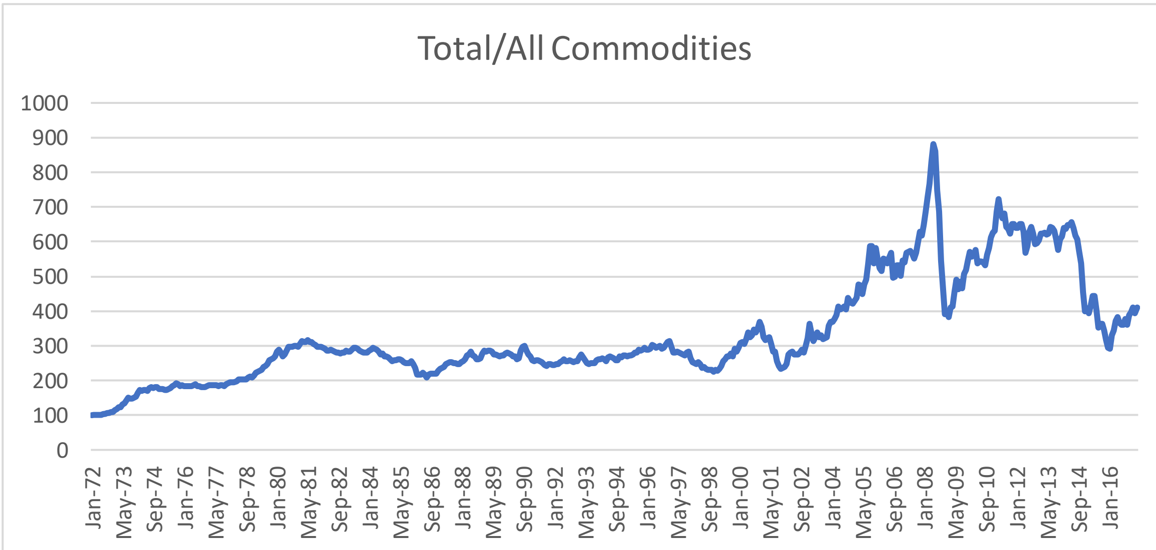
Variables	Mean	Median	Standard Deviation	Excess Kurtosis	Skewness
Total/All Commodities	333.13897	281.55	150.1932958	0.503875675	1.119653189
Total excluding energy	242.58988	222.9	72.65071075	-0.219761653	0.672084346
Energy	739.98511	570.3	492.4327296	0.752031483	1.154004097
Metals and Minerals	337.31488	272.45	167.2277264	-0.0791618	1.023304751
Agriculture	187.21691	176	45.99294214	1.220263	1.27240536
Fish	624.02536	629.25	367.8258373	-1.233922322	0.175864457
Forestry	261.47316	269.45	73.55742158	-0.931407814	-0.93140781
Canada - US Spot Exchange rate	1.221513	1.20463636	0.166674652	-0.828819931	0.254863707

Energy has the highest mean value of 739.98511. This implies that averagely Energy has the highest price rate followed by Fish, Metals and Minerals, all commodities, Forestry, Total excluding energy, Agriculture and Exchange rate having the least price rate.

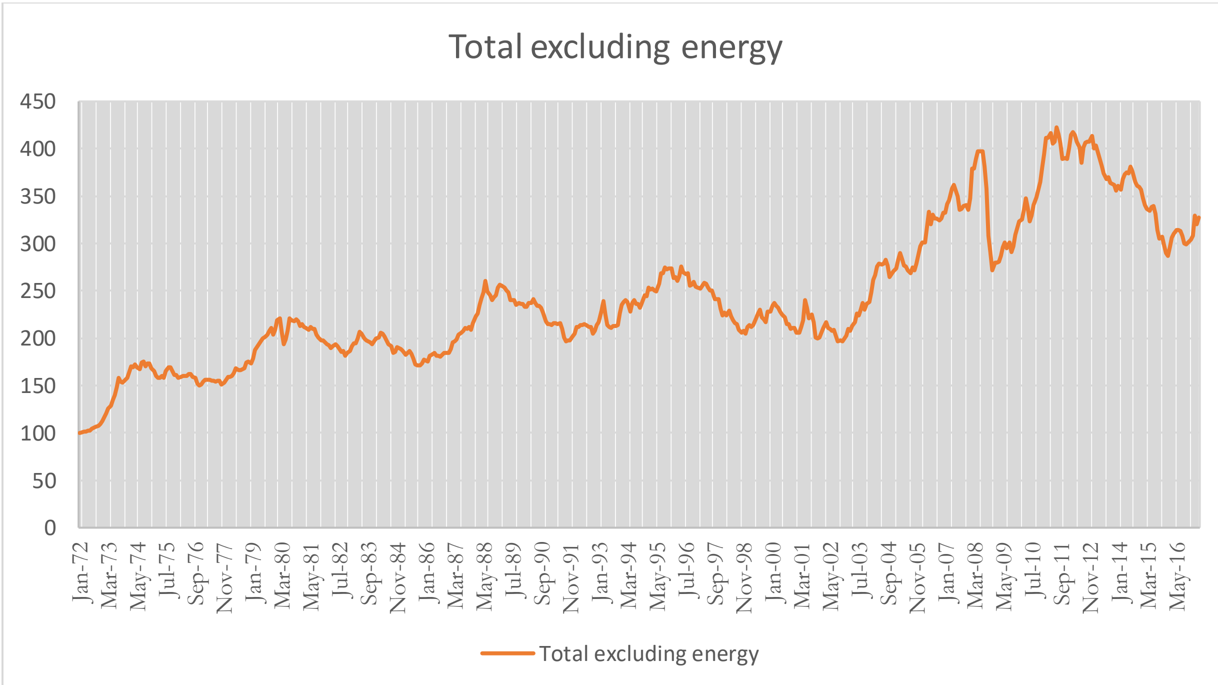
Energy has the highest standard deviation which means the values are farther away from its mean, hence variability, followed by Fish, Metals and Minerals, All commodities.

All the Variables had excess kurtosis less than 3, an indication that the variables were platykurtic and the distribution produces fewer and less extreme outliers than does the normal distribution.

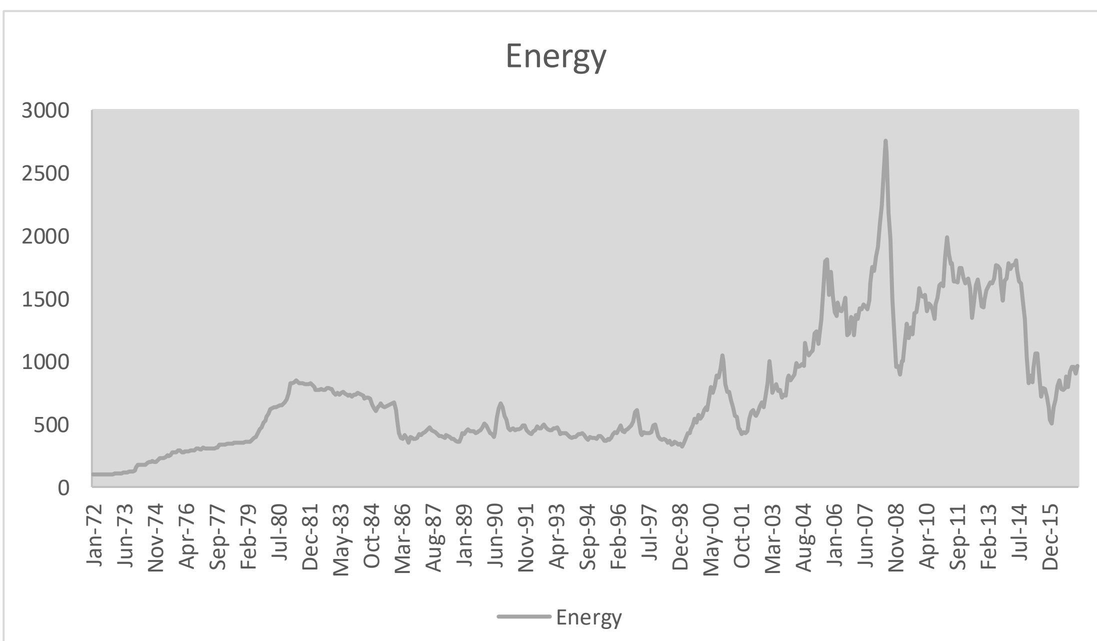
Forestry is negatively skewed which implies: the mode >median> mean,



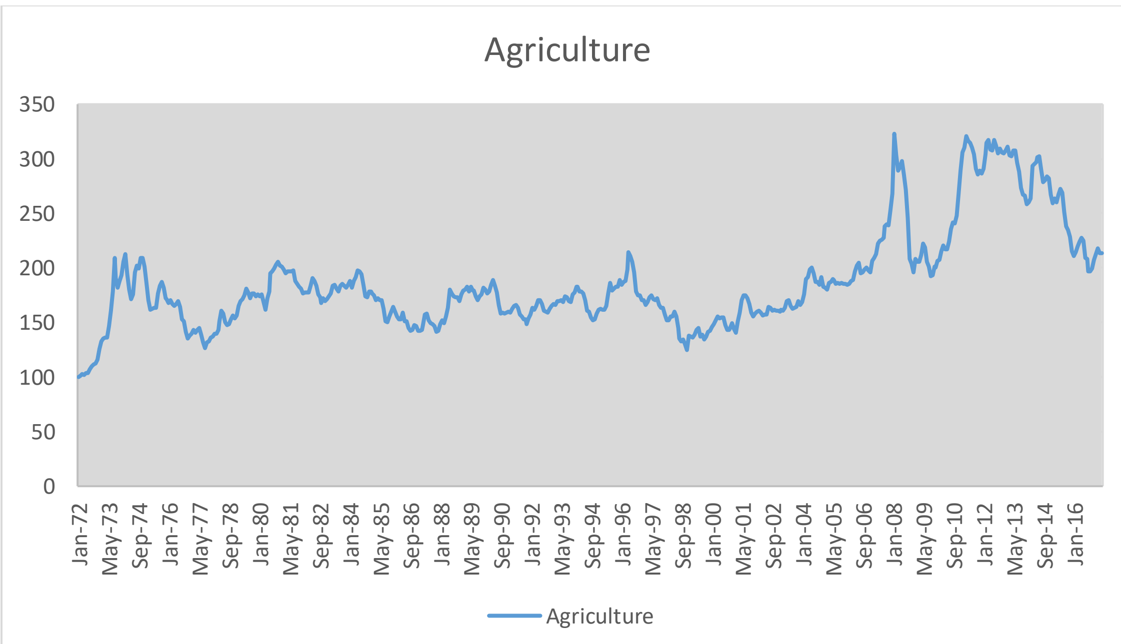
The figure above shows a clear view of the growth and decline of the Total/ All Commodities prices. Total /All commodities prices have shown a continuous increase in prices from 1972 to 2017. Prices jumped from 100 to 881.3 in Jan 2008 after which price fell and never got to that peak.



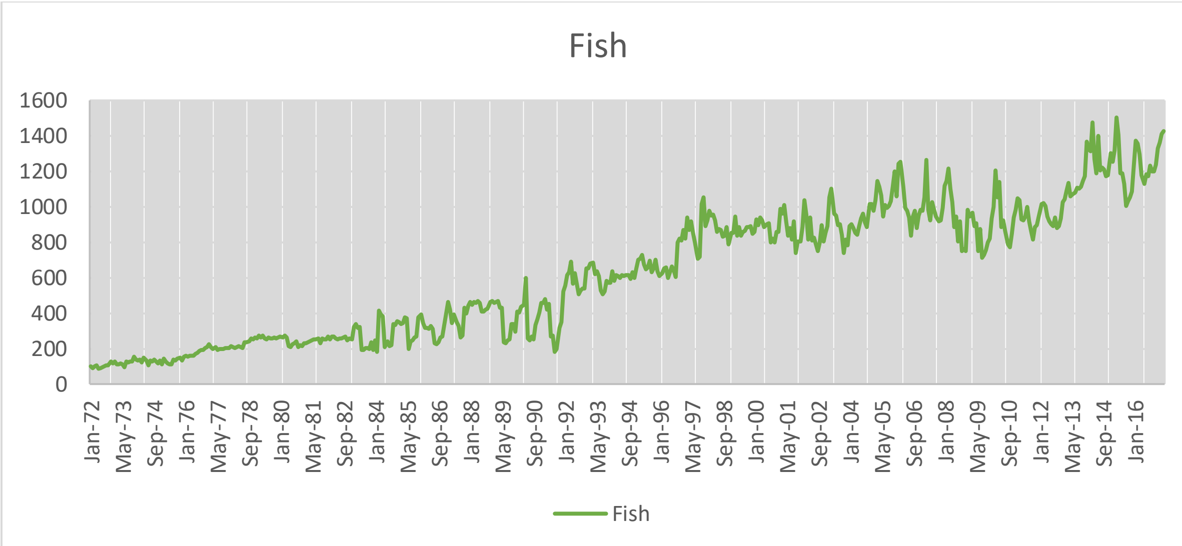
The figure above shows a clear view of the growth and decline of the Total Commodities excluding energy prices. Total commodities excluding energy prices have shown a fluctuation in price increases from 1972 to 2017. Prices jumped from 100 to 426 in 2011 and started dropping till they got to 327.7 in 2017.



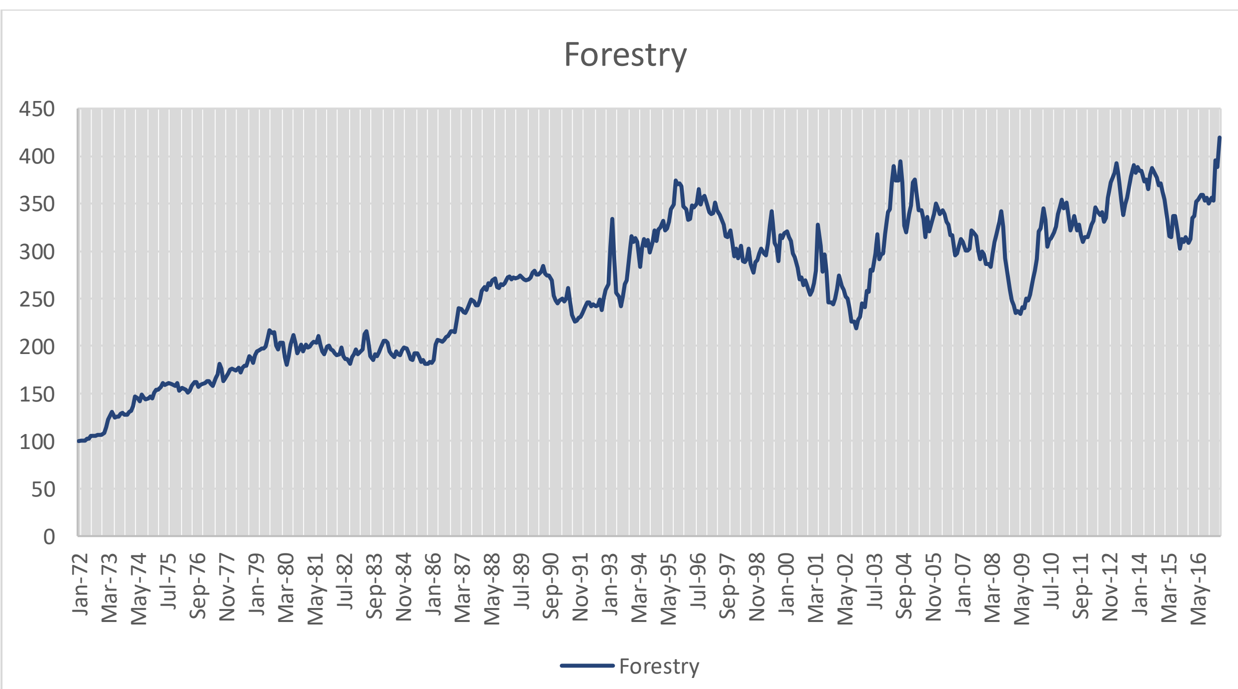
The figure below shows a clear view of the growth and decline of Energy prices. Energy prices have shown fluctuations in price increases from 1972 to 2017. Prices jumped from 100 to 822 in 1982 and started dropping till it got to 1050.90 in December 2000. Price manipulations continued from 2000 to 2017. The highest price of 2,755.80 was recorded in June 2008. The variability in the data affected the exchange rate and caused it to increase.



The figure above shows a clear view of the growth and decline of Agriculture prices. Agriculture prices have shown fluctuations in price increases from 1972 to 2017. Prices jumped from 100 to 322.9 in February 2008 and began to fall again.



The figure above shows a clear view of the growth and decline of Fish prices. Fish prices have shown fluctuations in price increases from 1972 to 2017. The highest price of 1503.60 was recorded in April 2015. Price manipulations continued from 2015 to 2017 and never got to that peak. The variability in the data affected the exchange rate and caused it to increase.



The figure above shows a clear view of the growth and decline of Forestry prices. Forestry prices have shown a steady increase in prices from 1972 to 2017. Prices increased from 100 to 419.5 in April 2017.

Multiple Linear Regression

This is a report on how I analyzed the effect of commodity prices on the Canada-US exchange rate using R Statistical Software. A multiple linear regression model was developed to determine the relationship. The Independent variable in this analysis is the commodity prices which are Total commodity, Total without Energy, Energy, Agriculture, Fish, Metals and Minerals and forestry whilst the dependent is the Canada-US exchange rate.

	Coefficients	*Standard Error*	*t Stat*	*P-value*
Intercept	1.2209	0.0279	43.777	3.92E-179
Total/All Commodities	-0.0079	0.0005	-15.013	8.78E-43
Total excluding energy	0.0189	0.0017	11.327	8.21E-27
Energy	0.0017	0.0001	14.761	1.27E-41
Metals and Minerals	-0.0037	0.0003	-11.049	1.04E-25
Agriculture	-0.0066	0.0005	-12.020	1.24E-29
Fish	0.0002	0.0000	8.205	1.72E-15
Forestry	-0.0031	0.0005	-6.530	1.53E-10

```
Residuals:
     Min        1Q    Median        3Q       Max
-0.186633 -0.058242 -0.004521 0.049349  0.304443


Residual standard error: 0.08943 on 536 degrees of freedom
Multiple R-squared:  0.7158,     Adjusted R-squared:  0.7121
F-statistic: 192.9 on 7 and 536 DF, p-value: < 2.2e-16
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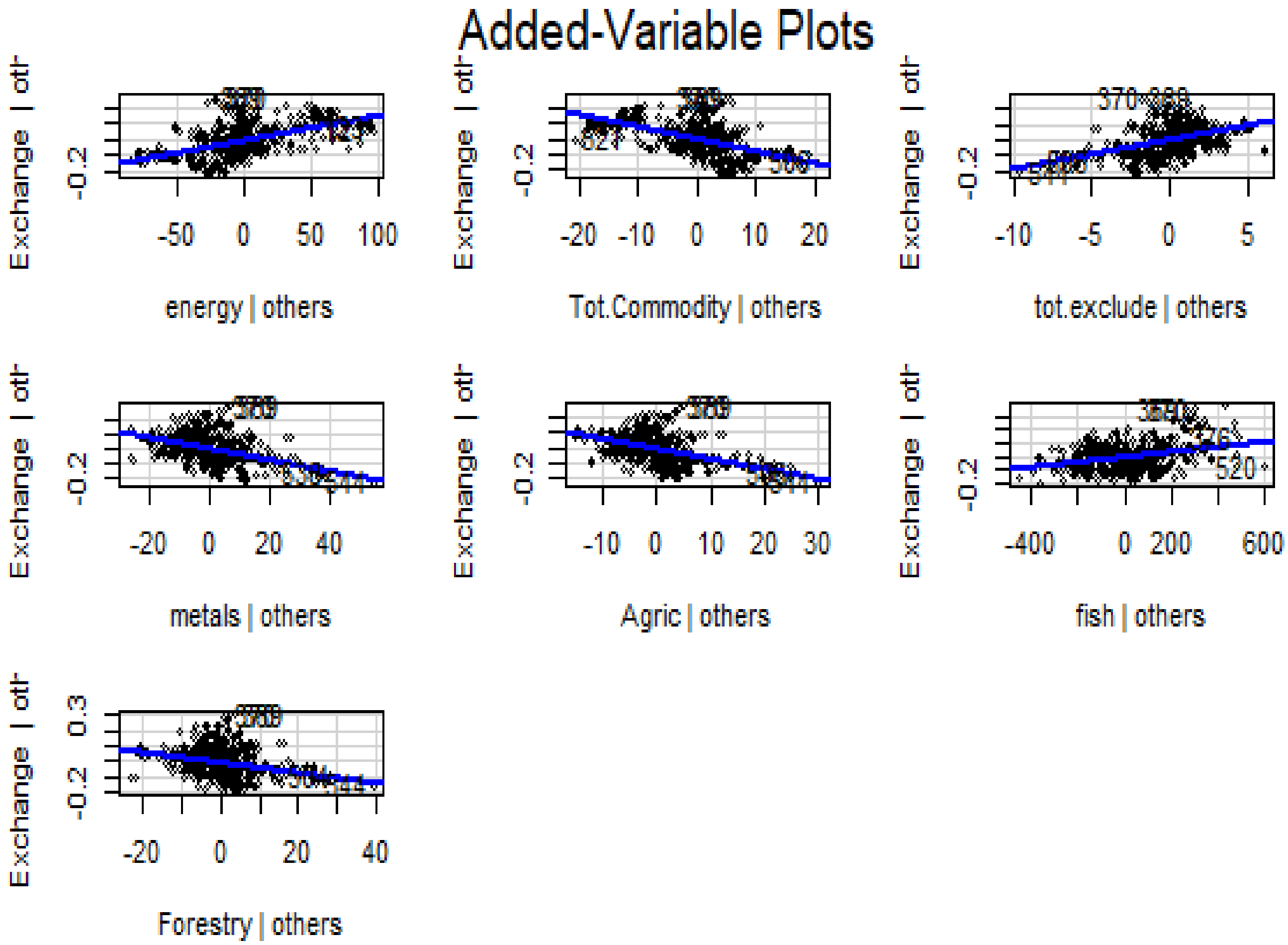


INTERPRETATION

It can be seen that the p-value of the F-statistic is < 2.2e-16, which is highly significant. This means that, at least, one of the predictor variables is significantly related to the outcome variable. To see which predictor variables are significant, you can examine the coefficients table, which shows the estimate of regression beta coefficients and the associated t-statistic p-values.

For a given predictor, the t-statistic evaluates whether or not there is a significant association between the predictor and the outcome variable, that is whether the beta coefficient of the predictor is significantly different from zero.

It can be seen that changes in Total excluding energy, Energy and Fish are significantly associated with changes in US- Canada exchange rate while changes in Total/All commodities, Metals and Minerals, Forestry and Agriculture budget are significantly associated with the Exchange rate.

For a given predictor variable, coefficient (b) can be interpreted as the average effect on y of a one-unit increase in the predictor, holding all other predictors fixed.

For example, the total excluding energy coefficient suggests that for every 200 dollars increase in Total excluding energy commodity holding all other predictors constant, we can expect an increase of 0.0189*200 = 3.78 exchange rate, on average.

We found that Metals and Minerals, Agriculture, and Forestry is not significant in the multiple regression model. This means that, for a fixed amount of Metals and Minerals, Agriculture and Forestry, there will not be significant changes in the exchange rate.

As the Metals and Minerals, Agriculture, and Forestry variable is not significant, it is possible to remove it from the model.

Model accuracy assessment

The overall quality of the model can be assessed by examining the R-squared (R^2) and Residual Standard Error (RSE).

R-squared

In multiple linear regression, the R2 represents the correlation coefficient between the observed values of the outcome variable (y) and the fitted (i.e., predicted) values of y. For this reason, the value of R will always be positive and will range from zero to one.

In our model, the adjusted R^2= 0.72, meaning that "72% of the variance in the measure of exchange rate can be predicted by Total commodity, Total without Energy, Energy, Agriculture, Fish, Metals and Minerals and forestry.

46% out of the 72% can be explained by Energy, Total excluding energy and Fish. These are the major variables that affect the exchange rate positively.

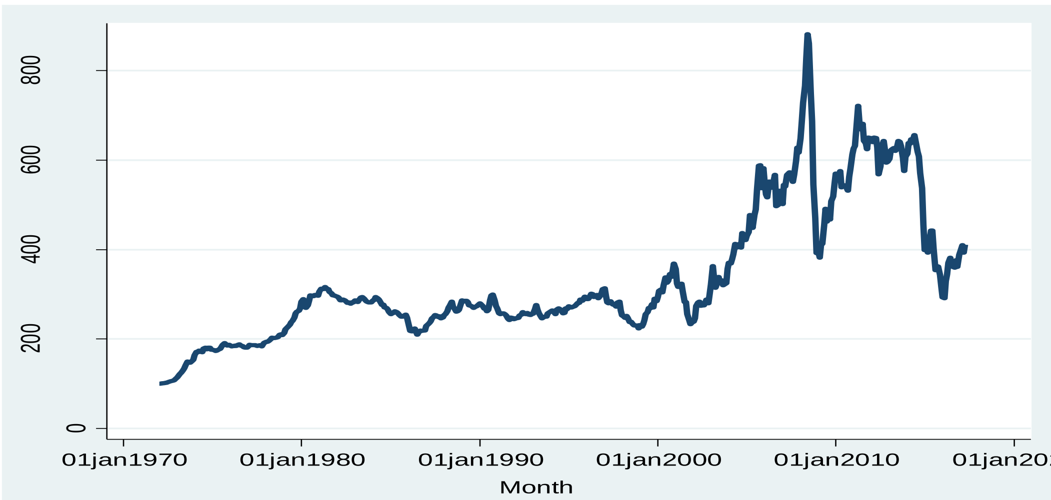
<u>RSE</u>

In our multiple regression example, the RSE is 0.08943 corresponding to a 7.3% error rate. Again, this is better.

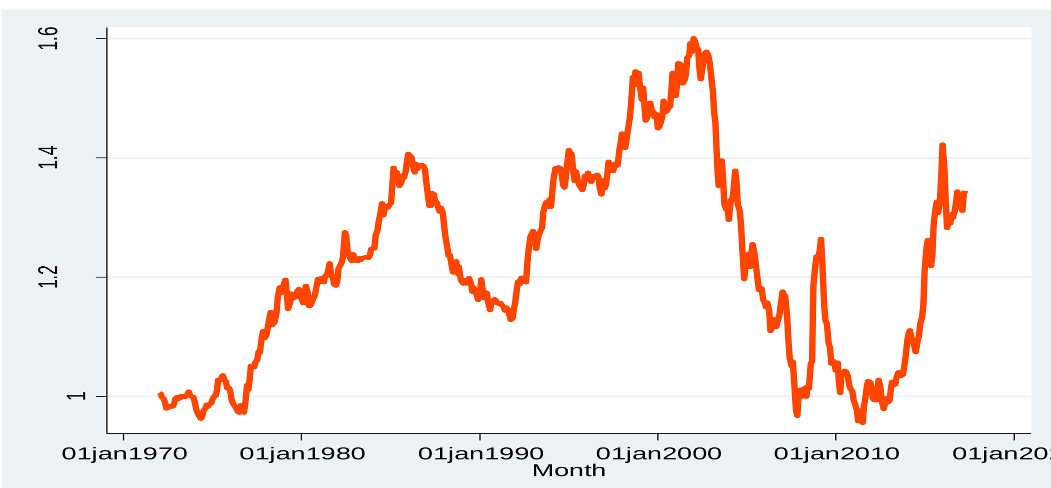
Empirical Analysis

First, I examine the relationship between the Canada – US exchange rate and the aggregate total commodity price index. The graph given below shows the trend plot of the total commodity prices and Canada – US exchange rates separately.

<u>Time series graph for All Commodities</u>



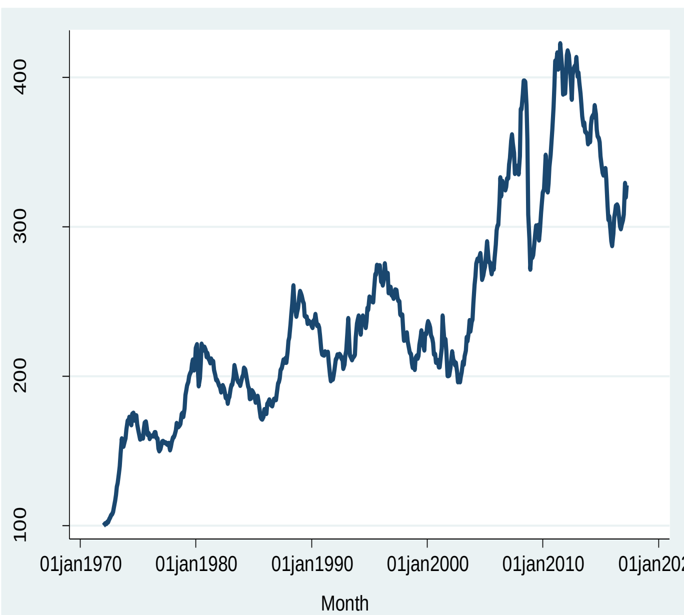
<u>Time series graph for Canada – US Exchange Rate</u>

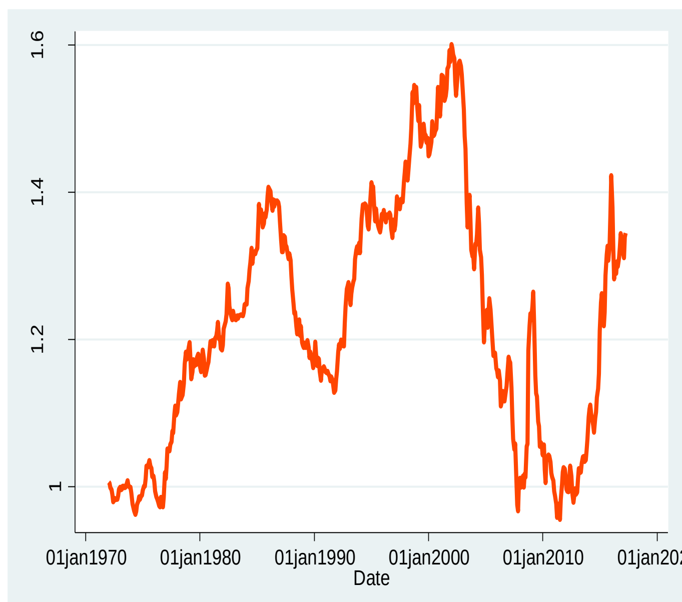


When we compare the two graphs, we can see that there appears to be a clear relationship between commodity prices and the exchange rate it appears to be a positive relationship between commodity price and exchange rate. The two diagrams above show random variation, and the mean keeps changing over time and this implies non-stationarity.

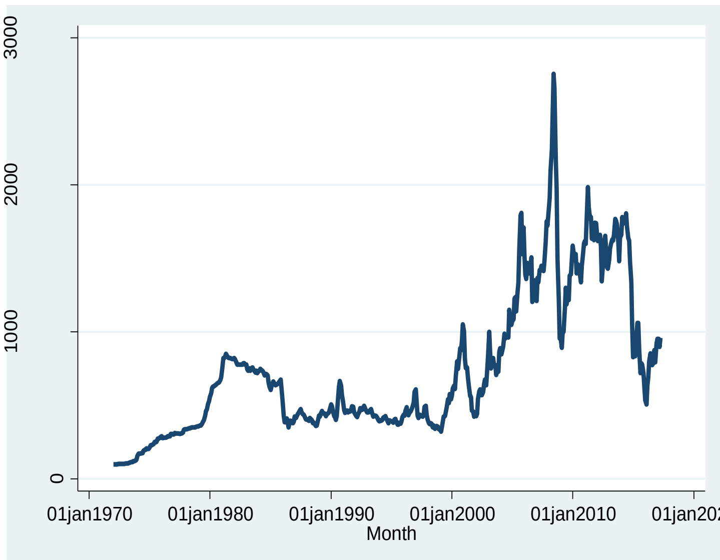
Second, I investigate the link between the Canadian-US exchange rate and other commodity prices. The graph below depicts a time-series graph of other commodity prices and Canadian-US exchange rates.

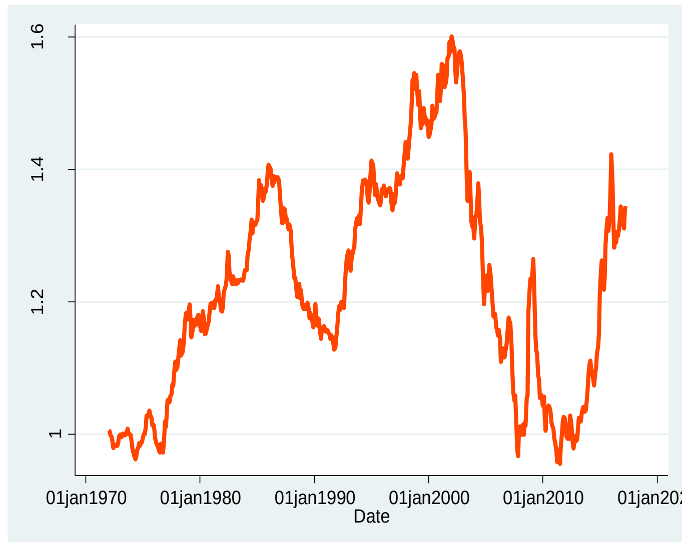
<u>Relationship Between Total Excluding Energy Prices and Canada – US Spot Exchange Rate</u>



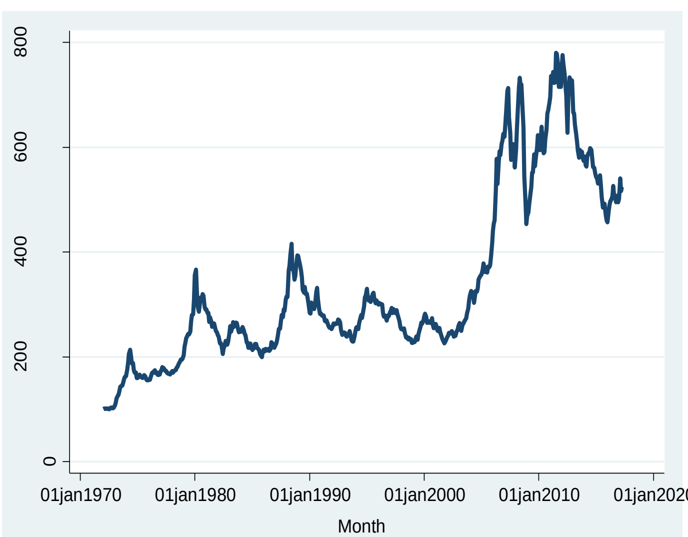


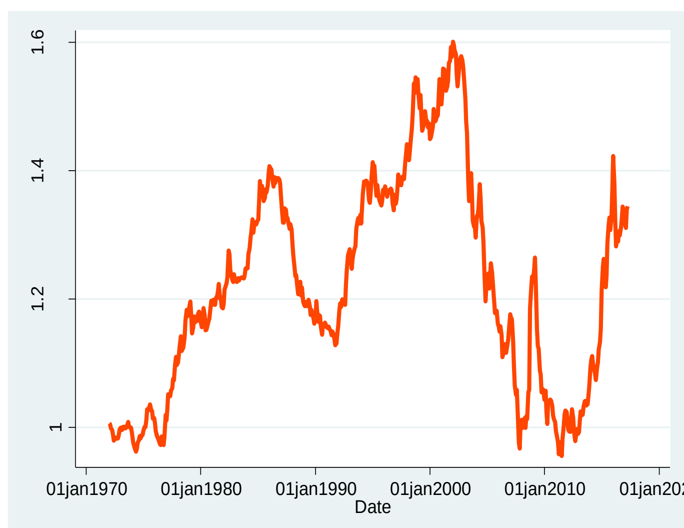
Relationship Between Total Excluding Energy Prices and Canada – US Spot Exchange Rate



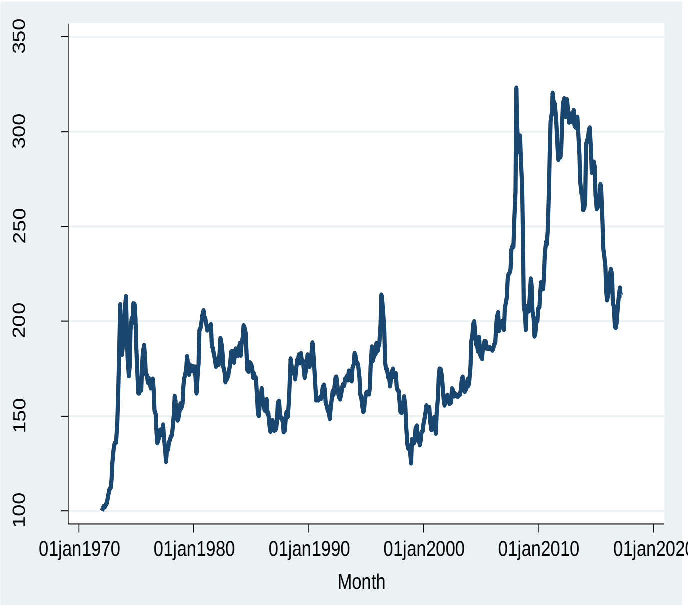


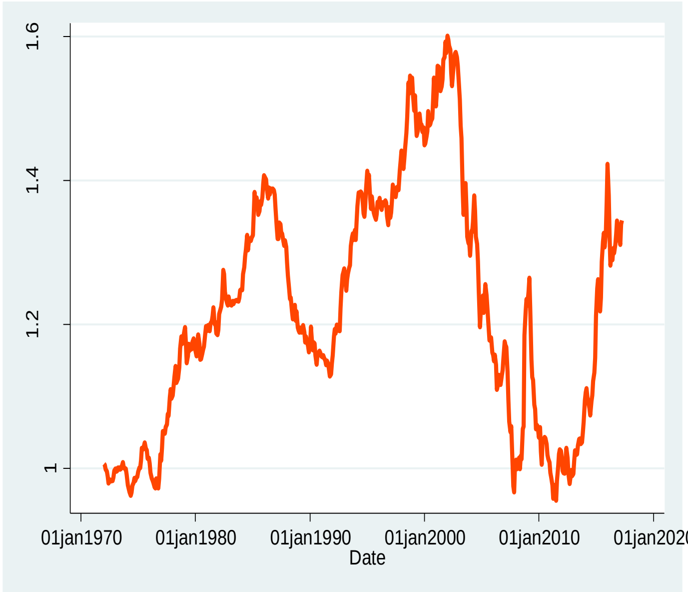
Relationship Between Metal & Minerals and Canada – US Spot Exchange Rate



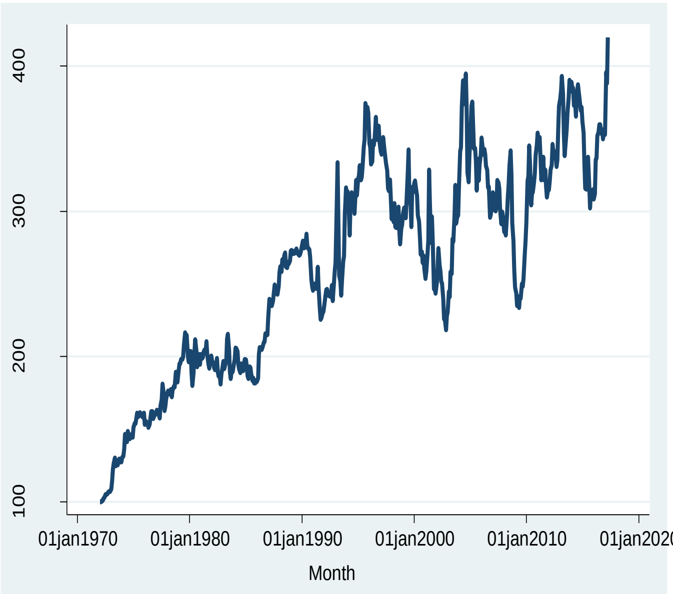


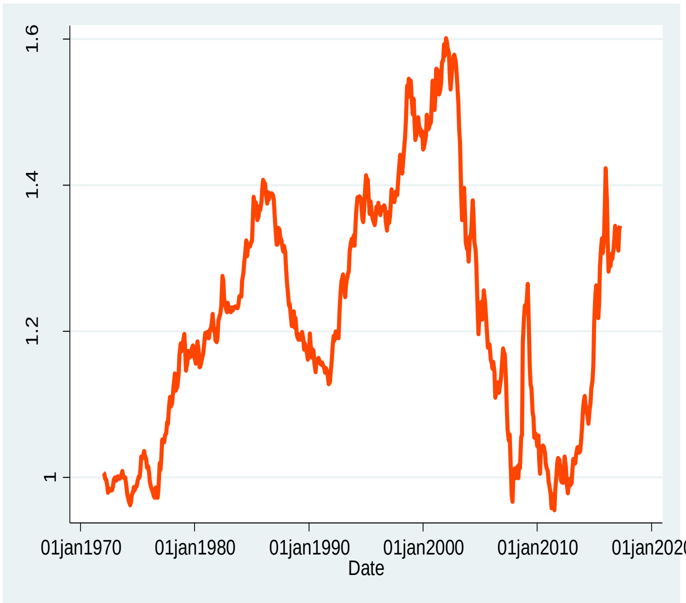
Relationship Between Agriculture and Canada – US Spot Exchange Rate



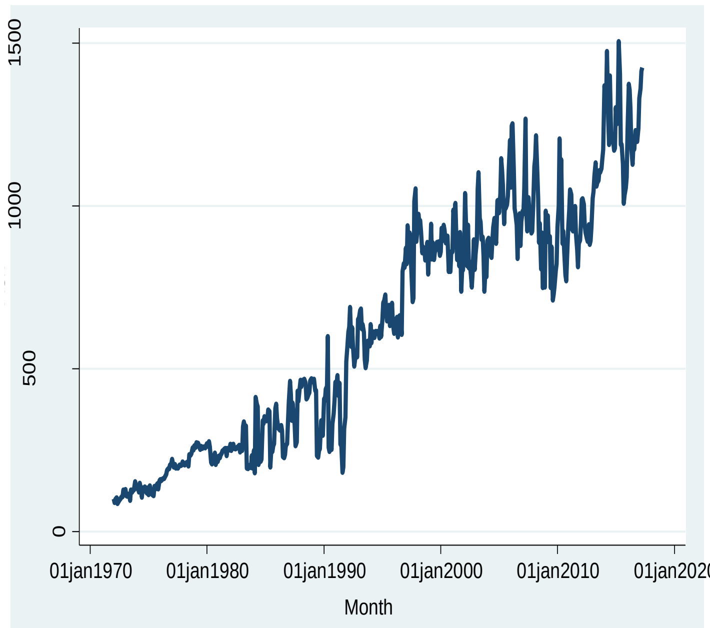


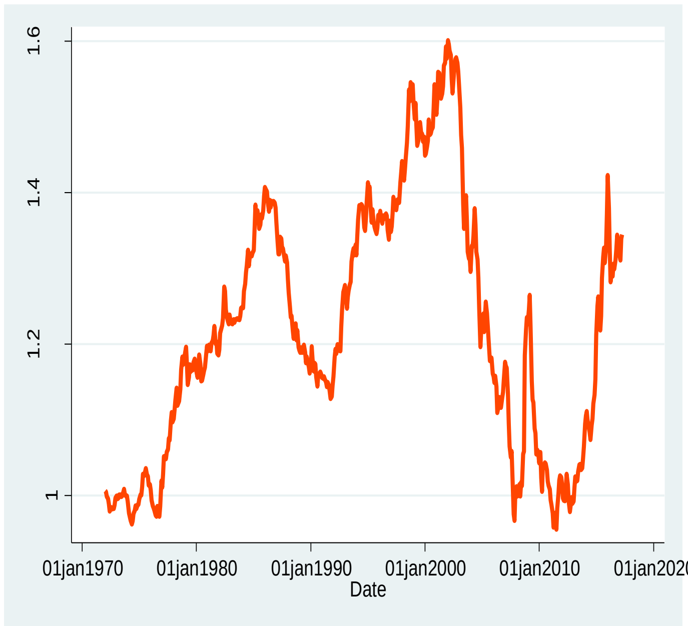
Relationship Between Forest and Canada – US Spot Exchange Rate





Relationship Between Forest and Canada – US Spot Exchange Rate





From the diagrams above, each of the time-series graphs above, we can easily observe that there appears to be a positive relationship between the Canada-US Spot Exchange Rate and any of these prices. This lends support to the hypothesis we derived by comparing the exchange rate to only the Total Commodity Prices. The diagrams above show random variation, and the mean keeps changing over time and this implies non-stationarity.

Unit Root Test.

Many economic and financial time series show trending or non-stationarity in the mean. Asset prices, exchange rates, and macroeconomic aggregates like real GDP are prime examples. Identifying the most appropriate form of the trend in the data is an important econometric task. If there is a data trend in the time series data, trend removal is necessary. Unit root tests can be used to determine whether a time series data has trending or non-stationarity. As a result, regression with non-stationary series is expected to produce spurious regression that is unsuitable for drawing correct statistical conclusions from the population. To test the relationship between the exchange rate and commodity prices, and to ensure that the results are valid without the persistence of shocks and that the regression is not spurious, the data must be stationary or trend stationery. So, for the purposes of this study, I would use the Augmented Dickey-Fuller (ADF) test to test for stationarity, because the ADF test is susceptible to the lag length chosen based on the sample size and looking at the nature of the data, it has a trend in it, so it is appropriate to use ADF with a trend and an intercept to test for stationarity. The ADF test is compatible with or without trend at both levels and first difference. The ADF is illustrated in the below equation as:

$$\triangle Y_t = \beta_0 + \beta_1 t + \beta Y_{t-1} + \sum_{j=1}^{p-1} Qj \triangle Y_{t-j}$$

Hypothesis

$H_{O:}$ β = 0{There is a Unit root and implies that data is non – Stationary}

$H_{1:}$ β < 0 {No Unit root and implies that data is Stationary}

The above equation Δ denotes the first difference in the exchange rate and price level, Yt is the time series being tested, t is the time trend variable, and k is the number of lags that are added to the model to ensure that the residuals, εt are white noise. The Akaike Information Criterion (AIC) is used to calculate the best lag length or k. The absence of rejection of the null hypothesis implies that the time series is non-stationary, whereas the rejection of the null hypothesis indicates that the time series is stationary.

Unit Root Testing for Canada – US Spot Exchange Rate

First, I looked for the unit root for the Canada-US exchange rate, taking both with and without trend into account. I used the Augmented Dicker-Fuller (ADF), test to determine if there is a unit root in the data. I determined the Lag order prior to the ADF test. Because AIC yields the optimal lag of 2 (See Appendix 2), I set the lag to 2 and proceed with implementing the Augmented Dickey-Fuller (ADF) Test. In a reference form (Appendix 3), the test statistic is (-1.786) and the 5 % critical value is (-2.860). But the Stata output shows that the test statistic's absolute value of 1.786 is less than the 5% critical value of 2.860. Including the trend, yields the same result as shown in (Appendix 5), the test statistic value of 1.751 against the critical value of 3.410. As a result, the null hypothesis cannot be rejected, implying that there is enough evidence to conclude that the Canada-US Spot Exchange rate has a unit root and hence, non – stationary.

In my analysis, I also used the DF-GLS Test to look for a unit root in the Canada-US exchange rate. The results show that all computed test statistics are less than the critical values, so we cannot reject the null hypothesis, implying that there is a unit root in the data, and this condition holds for with and without trend, as shown in appendices 6 and 7.

Finally, I used the KPSS test to check for unit roots. The critical value from the KPSS result (see appendix 8) is 0.146 (5% level), which is less than the test statistic value of 0.446. As a result, I reject the null hypothesis that it is not trend stationery and conclude that there is a unit root in the Canada-US spot exchange rate from 1972 to 2020. The presence of a unit root in the Canada-US spot exchange rate as confirmed by the DF-GLS test. (Appendices 8 and 9)

Unit Root Testing for All Commodities Prices

Similarly, I check to see if the Total Commodity price reflects the unit root process. I determined the lag order and included it in the ADF test. The AIC yields the optimal lag of 2 (See Appendix 11), therefore I set the lag to 2 and proceed with implementing the Augmented Dickey-Fuller (ADF) Test. The test statistic is 2.398 (Absolute value) and the critical value is 2.860 (5% level of significant, absolute value), we reject the null hypothesis and conclude that the process has a unit root because the critical value exceeds the test statistic (See Appendix 12). Including the trend, yields the same result as shown in (Appendix 13), the test statistic value of 3.399 against the critical value of 3.960. As a result, the null hypothesis cannot be rejected, implying that there is enough evidence to conclude that the total commodities Price has a unit root and hence, non – stationary.

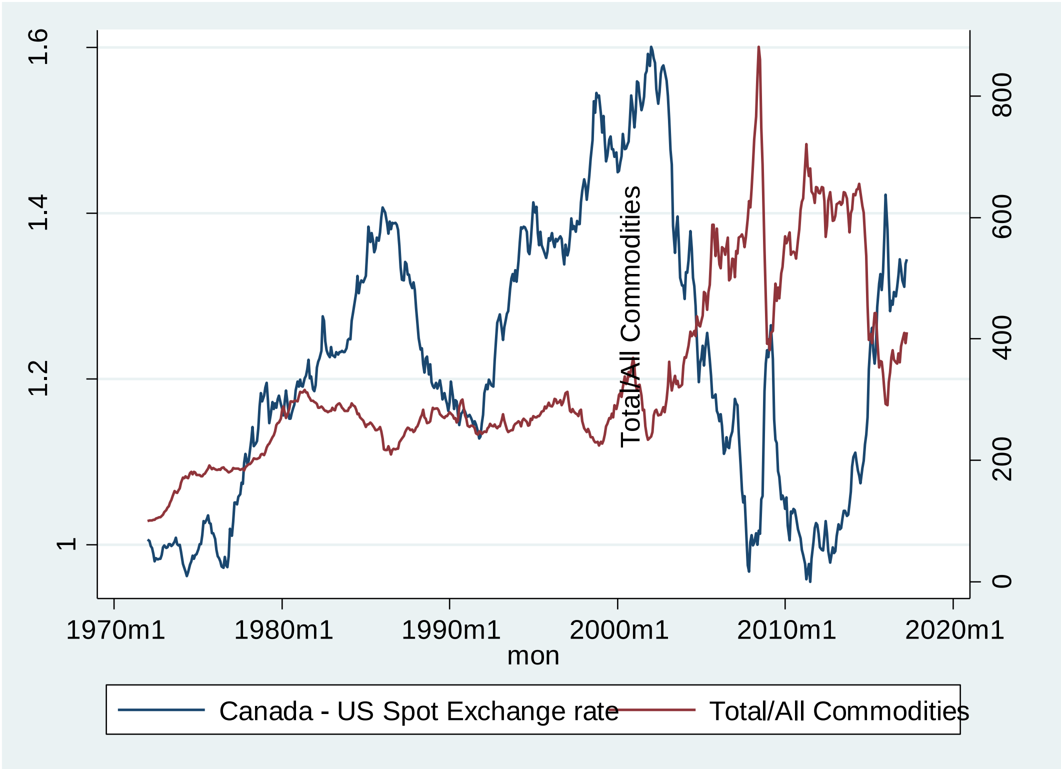
I also used the DF-GLS Test to look for a unit root in the total commodities Price in my analysis. The results show that computed test statistics are less than the critical values, so we cannot reject the null hypothesis, implying that there is a unit root in the data, as shown in appendices 14 and 15.

Lastly, I checked for unit roots using the KPSS test. The critical value from the KPSS result is 0.146 (5 per cent level), which is less than the test statistic value of 0.255. As a result, I reject the null hypothesis that it is not trend stationery and conclude that the Canada-US spot exchange rate has a unit root from 1972 to 2020. The DF-GLS test confirmed the presence of a unit root in the Total commodities prices (See Appendix 16 and 17).

Testing for Cointegration

Two variables are said to be cointegrated if their linear combination is integrated of order 0. As a result, even if there are unit root processes in the data of two independent variables, the regression of one variable on the other will no longer be spurious. The regressions would still be spurious if the variables were not cointegrated. Cointegration tests identify scenarios in which two or more non-stationary time series are integrated together in such a way that they cannot deviate from equilibrium over time. The tests are used to determine the sensitivity of two variables to the same average price over a given time. The graph below depicts the graphical relationship between the Canada-US spot exchange rate and total commodity price from 1972 to 2017. According to the diagram below, the two diagrams move together and exhibit similar behaviours, implying that these variables are likely to be cointegrated. In order to test for cointegration, I conducted the test using Stata shows that the AIC has an optimal lag is 2 (Appendix 18) and I set the lag to 2 and proceed with implementing the Augmented Dickey-Fuller (ADF) Test. The test statistic in ADF showed a value of -2.96 and the asymptotic critical value of -3.80 at a 5% level of significance. Hence, we fail to reject the null hypothesis and conclude that is unit root and therefore, Canada-US spot exchange rate and total commodity are not cointegrated. There is a negative relation between the Canada-US spot exchange rate and total commodity. (See Appendix 20).

<u>Canada-US spot exchange rate and Total Commodity Price</u>



Conclusion

This paper conducted empirical research on the relationship between the Canadian-US exchange rate and Canadian commodity prices. The study's findings indicate the presence of unit root in commodity prices as well as exchange rates. According to the study's findings, there is a negative relationship between the Canadian-US exchange rate and total commodity prices. As a result, a 1% increase in total commodity prices will result in a 0.123 per cent depreciation in the Canada-US exchange rate. The study's findings, however, are consistent with the literature.

APPENDIX

Appendix 1: Dickey-Fuller test of Unit root for the Canada-US exchange rate (Without Trend)

```
Dickey-Fuller test for unit root                      Number of obs   =       543

                               ---------- Interpolated Dickey-Fuller ---------
                  Test         1% Critical       5% Critical      10% Critical
               Statistic           Value             Value            Value
------------------------------------------------------------------------------
 Z(t)             -1.470            -3.430            -2.860            -2.570
------------------------------------------------------------------------------
MacKinnon approximate p-value for Z(t) = 0.5485

------------------------------------------------------------------------------
       D.ner |      Coef.   Std. Err.      t    P>|t|     [95% Conf. Interval]
-------------+----------------------------------------------------------------
         ner |
         L1. |  -.0067173   .0045709    -1.47   0.142    -.0156963    .0022616
             |
       _cons |   .0088271   .0056341     1.57   0.118    -.0022403    .0198945
------------------------------------------------------------------------------
```

Appendix 2: Selecting a lag order for the Canada-US exchange rate (Without Trend)

```
.  varsoc ner

   Selection-order criteria
   Sample:  1972m5 - 2017m4                     Number of obs      =       540
  +---------------------------------------------------------------------------+
  |lag |    LL      LR      df    p      FPE       AIC      HQIC      SBIC    |
  |----+----------------------------------------------------------------------|
  |  0 |  203.329                      .027674  -.749367  -.746259   -.74142  |
  |  1 |  1410.56  2414.5     1  0.000  .000318  -5.21688  -5.21066  -5.20099  |
  |  2 |  1434.39   47.67*    1  0.000  .000292* -5.30146* -5.29213* -5.27761* |
  |  3 |  1435.19  1.5895     1  0.207  .000292   -5.3007  -5.28826  -5.26891  |
  |  4 |  1435.25  .12616     1  0.722  .000293  -5.29723  -5.28168  -5.25749  |
  +---------------------------------------------------------------------------+
   Endogenous:  ner
    Exogenous:  _cons
```

Appendix 3: Augmented Dickey-Fuller test of Unit root Canada-US exchange rate (Without Trend)

```
.  dfuller ner, lag (2) regress

Augmented Dickey-Fuller test for unit root          Number of obs   =       541

                              ---------- Interpolated Dickey-Fuller ---------
                  Test          1% Critical        5% Critical      10% Critical
               Statistic           Value              Value             Value
------------------------------------------------------------------------------
 Z(t)             -1.786            -3.430             -2.860            -2.570
------------------------------------------------------------------------------
MacKinnon approximate p-value for Z(t) = 0.3874

------------------------------------------------------------------------------
       D.ner |      Coef.   Std. Err.      t    P>|t|     [95% Conf. Interval]
-------------+----------------------------------------------------------------
         ner |
         L1. |  -.0078739   .0044085    -1.79   0.075    -.0165339    .0007861
         LD. |   .3058534   .0429581     7.12   0.000     .2214668    .3902399
        L2D. |   -.054285   .0431228    -1.26   0.209    -.1389951    .0304252
             |
       _cons |   .0101039   .0054349     1.86   0.064    -.0005725    .0207802
------------------------------------------------------------------------------
```

Appendix 4: Selecting a lag order for the Canada-US exchange rate (With Trend)

```
.  **with trend**
.  varsoc ner, exo(mon)

   Selection-order criteria
   Sample:  1972m5 - 2017m4                      Number of obs      =       540
  +---------------------------------------------------------------------------+
  |lag |    LL      LR      df    p      FPE       AIC       HQIC      SBIC    |
  |----+----------------------------------------------------------------------|
  |  0 |  210.993                          .027  -.774049  -.767833  -.758154  |
  |  1 |  1410.58  2399.2    1  0.000  .000319  -5.21324  -5.20392   -5.1894  |
  |  2 |  1434.39  47.634*   1  0.000  .000293* -5.29775* -5.28532* -5.26596* |
  |  3 |  1435.19  1.5916    1  0.207  .000293    -5.297  -5.28146  -5.25726  |
  |  4 |  1435.25  .12713    1  0.721  .000294  -5.29353  -5.27488  -5.24584  |
  +---------------------------------------------------------------------------+
   Endogenous:  ner
    Exogenous:  mon  _cons
```

Appendix 5: Augmented Dickey-Fuller test of Unit root for Canada-US exchange rate (With Trend)

```
.  dfuller ner, lag (2) trend regress

Augmented Dickey-Fuller test for unit root         Number of obs   =       541

                               ---------- Interpolated Dickey-Fuller ---------
                  Test         1% Critical       5% Critical      10% Critical
               Statistic           Value             Value             Value
------------------------------------------------------------------------------
 Z(t)             -1.751            -3.960            -3.410            -3.120
------------------------------------------------------------------------------
MacKinnon approximate p-value for Z(t) = 0.7279

------------------------------------------------------------------------------
D.ner         |      Coef.   Std. Err.      t    P>|t|     [95% Conf. Interval]
--------------+---------------------------------------------------------------
          ner |
          L1. |  -.0078477   .0044819    -1.75   0.081    -.0166518    .0009565
          LD. |   .3058298   .0430039     7.11   0.000     .2213529    .3903066
         L2D. |  -.0543277   .0431819    -1.26   0.209    -.1391541    .0304988
       _trend |  -1.59e-07   4.76e-06    -0.03   0.973    -9.51e-06    9.20e-06
        _cons |   .0101153   .0054507     1.86   0.064    -.0005921    .0208227
------------------------------------------------------------------------------
```

Appendix 6: DF-GLS Test for the Canada-US exchange rate (Without Trend)

```
.  ***DF-GLS with no trend**
.  dfgls ner, ers notrend

DF-GLS for ner                                          Number of obs =   525
Maxlag = 18 chosen by Schwert criterion

               DF-GLS mu       1% Critical       5% Critical      10% Critical
  [lags]     Test Statistic        Value             Value             Value
------------------------------------------------------------------------------
    18          -0.825            -2.580            -1.950            -1.620
    17          -0.686            -2.580            -1.950            -1.620
    16          -0.652            -2.580            -1.950            -1.620
    15          -0.639            -2.580            -1.950            -1.620
    14          -0.720            -2.580            -1.950            -1.620
    13          -0.736            -2.580            -1.950            -1.620
    12          -0.830            -2.580            -1.950            -1.620
    11          -0.947            -2.580            -1.950            -1.620
    10          -0.861            -2.580            -1.950            -1.620
    9           -0.660            -2.580            -1.950            -1.620
    8           -0.660            -2.580            -1.950            -1.620
    7           -0.638            -2.580            -1.950            -1.620
    6           -0.721            -2.580            -1.950            -1.620
    5           -0.765            -2.580            -1.950            -1.620
    4           -0.789            -2.580            -1.950            -1.620
    3           -0.620            -2.580            -1.950            -1.620
    2           -0.645            -2.580            -1.950            -1.620
    1           -0.721            -2.580            -1.950            -1.620

Opt Lag (Ng-Perron seq t) = 18 with RMSE  .0166195
Min SC   = -8.093976 at lag  1 with RMSE  .0172677
Min MAIC = -8.128596 at lag 13 with RMSE  .0167364
```

Appendix 7: DF-GLS Test for the Canada-US exchange rate (With Trend)

```
.  ***DF_GLS test with trend***
.  dfgls ner, ers

DF-GLS for ner                                          Number of obs =   525
Maxlag = 18 chosen by Schwert criterion

               DF-GLS tau      1% Critical       5% Critical      10% Critical
  [lags]     Test Statistic        Value             Value             Value
------------------------------------------------------------------------------
    18           -1.673           -3.480            -2.890            -2.570
    17           -1.508           -3.480            -2.890            -2.570
    16           -1.470           -3.480            -2.890            -2.570
    15           -1.457           -3.480            -2.890            -2.570
    14           -1.533           -3.480            -2.890            -2.570
    13           -1.548           -3.480            -2.890            -2.570
    12           -1.650           -3.480            -2.890            -2.570
    11           -1.793           -3.480            -2.890            -2.570
    10           -1.687           -3.480            -2.890            -2.570
    9            -1.446           -3.480            -2.890            -2.570
    8            -1.446           -3.480            -2.890            -2.570
    7            -1.421           -3.480            -2.890            -2.570
    6            -1.513           -3.480            -2.890            -2.570
    5            -1.562           -3.480            -2.890            -2.570
    4            -1.589           -3.480            -2.890            -2.570
    3            -1.407           -3.480            -2.890            -2.570
    2            -1.434           -3.480            -2.890            -2.570
    1            -1.520           -3.480            -2.890            -2.570

Opt Lag (Ng-Perron seq t) = 18 with RMSE  .0165792
Min SC   = -8.098129 at lag  1 with RMSE  .0172319
Min MAIC = -8.125237 at lag 13 with RMSE     .0167
```

Appendix 8: KPSS Test for the Canada-US exchange rate (With Trend)

```
.  kpss ner, auto //with trend

KPSS test for ner

Automatic bandwidth selection (maxlag) = 17
Autocovariances weighted by Bartlett kernel

Critical values for H0: ner is trend stationary

10%: 0.119  5% : 0.146  2.5%: 0.176  1% : 0.216

Lag order    Test statistic
   17             .446
```

Appendix 9: KPSS Test for the Canada-US exchange rate (Without Trend)

```
.  kpss ner, notrend auto //with no trend

KPSS test for ner

Automatic bandwidth selection (maxlag) = 17
Autocovariances weighted by Bartlett kernel

Critical values for H0: ner is level stationary

10%: 0.347  5% : 0.463  2.5%: 0.574  1% : 0.739

Lag order    Test statistic
   17            .527
```

Unit root test results for Total Commodities Price

Appendix 10: Dickey-Fuller test of unit root for Total Commodities Price (Without Trend)

```
Dickey-Fuller test for unit root                   Number of obs   =       543

                               ---------- Interpolated Dickey-Fuller ---------
                  Test         1% Critical       5% Critical      10% Critical
               Statistic           Value             Value             Value
------------------------------------------------------------------------------
 Z(t)             -1.772            -3.430            -2.860            -2.570
------------------------------------------------------------------------------
MacKinnon approximate p-value for Z(t) = 0.3942

------------------------------------------------------------------------------
      D.tcom |      Coef.   Std. Err.      t    P>|t|     [95% Conf. Interval]
-------------+----------------------------------------------------------------
        tcom |
         L1. |  -.0090126   .0050849    -1.77   0.077    -.0190012    .0009761
             |
       _cons |   3.573345   1.857454     1.92   0.055    -.0753608     7.22205
------------------------------------------------------------------------------
```

Appendix 11: Lag order selection for Total Commodity Prices (Without Trend)

```
.  varsoc tcom

   Selection-order criteria
   Sample:  1972m5 - 2017m4                     Number of obs      =       540
```

lag	LL	LR	df	p	FPE	AIC	HQIC	SBIC
0	-3469.32				22362.8	12.853	12.8561	12.861
1	-2321.23	2296.2	1	0.000	319.477	8.60456	8.61078	8.62046
2	-2284.23	74.003*	1	0.000	279.596	8.47122	8.48055*	8.49506*
3	-2282.77	2.9264	1	0.087	279.117*	8.46951*	8.48194	8.5013
4	-2282.67	.18713	1	0.665	280.055	8.47286	8.48841	8.5126

```
   Endogenous:  tcom
    Exogenous:  _cons
```

Appendix 12: Augmented Dickey-Fuller test of Unit root for Total Commodities Price (Without Trend)

```
Augmented Dickey-Fuller test for unit root         Number of obs   =       541
```

	Test Statistic	1% Critical Value	5% Critical Value	10% Critical Value
		Interpolated Dickey-Fuller		
Z(t)	-2.398	-3.430	-2.860	-2.570

```
MacKinnon approximate p-value for Z(t) = 0.1422
```

D.tcom	Coef.	Std. Err.	t	P>|t|	[95% Conf.	Interval]
tcom						
L1.	-.0114919	.0047919	-2.40	0.017	-.020905	-.0020787
LD.	.3314978	.0429171	7.72	0.000	.2471919	.4158037
L2D.	.073547	.0430771	1.71	0.088	-.0110734	.1581673
_cons	4.187676	1.75056	2.39	0.017	.7488915	7.62646

Appendix 13: Lag order selection for Total Commodity Prices (With Trend)

```
. varsoc tcom, exo(mon)

  Selection-order criteria
  Sample:  1972m5 - 2017m4                    Number of obs      =       540
```

lag	LL	LR	df	p	FPE	AIC	HQIC	SBIC
0	-3207.53				8512.33	11.8871	11.8934	11.903
1	-2320.48	1774.1	1	0.000	319.772	8.60549	8.61481	8.62933
2	-2281.66	77.639	1	0.000	277.977	8.46541	8.47785	8.4972*
3	-2279.61	4.1096*	1	0.043	276.893*	8.46151*	8.47705*	8.50124
4	-2279.6	.00661	1	0.935	277.917	8.4652	8.48385	8.51288

```
  Endogenous:  tcom
   Exogenous:  mon  _cons
```

Appendix 13: Augmented Dickey-Fuller test of Unit root for Total Commodities Price (With Trend)

```
Augmented Dickey-Fuller test for unit root          Number of obs   =       540
```

	Test Statistic	Interpolated Dickey-Fuller 1% Critical Value	5% Critical Value	10% Critical Value
Z(t)	-3.399	-3.960	-3.410	-3.120

```
MacKinnon approximate p-value for Z(t) = 0.0516
```

D.tcom	Coef.	Std. Err.	t	P>|t|	[95% Conf.	Interval]
tcom						
L1.	-.0272272	.0080111	-3.40	0.001	-.0429643	-.01149
LD.	.3375174	.042912	7.87	0.000	.2532203	.4218144
L2D.	.0885498	.0453673	1.95	0.051	-.0005704	.17767
L3D.	-.0035193	.0435339	-0.08	0.936	-.0890381	.0819994
_trend	.0188766	.0076392	2.47	0.014	.00387	.0338833
_cons	4.281027	1.755074	2.44	0.015	.8333306	7.728723

Appendix 14: DF GLS Test for Total Commodities with Trend

```
.  dfgls tcom, ers

DF-GLS for tcom                                          Number of obs =   525
Maxlag = 18 chosen by Schwert criterion

               DF-GLS tau      1% Critical       5% Critical      10% Critical
  [lags]     Test Statistic       Value             Value             Value
------------------------------------------------------------------------------
    18          -1.993           -3.480            -2.890            -2.570
    17          -2.312           -3.480            -2.890            -2.570
    16          -2.318           -3.480            -2.890            -2.570
    15          -2.132           -3.480            -2.890            -2.570
    14          -2.276           -3.480            -2.890            -2.570
    13          -2.227           -3.480            -2.890            -2.570
    12          -2.460           -3.480            -2.890            -2.570
    11          -2.671           -3.480            -2.890            -2.570
    10          -2.466           -3.480            -2.890            -2.570
    9           -2.351           -3.480            -2.890            -2.570
    8           -2.505           -3.480            -2.890            -2.570
    7           -2.602           -3.480            -2.890            -2.570
    6           -2.676           -3.480            -2.890            -2.570
    5           -2.785           -3.480            -2.890            -2.570
    4           -3.386           -3.480            -2.890            -2.570
    3           -3.304           -3.480            -2.890            -2.570
    2           -3.363           -3.480            -2.890            -2.570
    1           -3.105           -3.480            -2.890            -2.570

Opt Lag (Ng-Perron seq t) = 18 with RMSE  16.17363
Min SC   =  5.665195 at lag  1 with RMSE  16.78805
Min MAIC =  5.657157 at lag 18 with RMSE  16.17363
```

Appendix 15: DF GLS Test for Total Commodities Price (Without Trend)

```
.  dfgls tcom, ers notrend

DF-GLS for tcom                                          Number of obs =   525
Maxlag = 18 chosen by Schwert criterion

               DF-GLS mu       1% Critical       5% Critical      10% Critical
  [lags]     Test Statistic       Value             Value             Value
------------------------------------------------------------------------------
    18          -0.341           -2.580            -1.950            -1.620
    17          -0.471           -2.580            -1.950            -1.620
    16          -0.479           -2.580            -1.950            -1.620
    15          -0.418           -2.580            -1.950            -1.620
    14          -0.470           -2.580            -1.950            -1.620
    13          -0.461           -2.580            -1.950            -1.620
    12          -0.545           -2.580            -1.950            -1.620
    11          -0.626           -2.580            -1.950            -1.620
    10          -0.554           -2.580            -1.950            -1.620
    9           -0.512           -2.580            -1.950            -1.620
    8           -0.580           -2.580            -1.950            -1.620
    7           -0.625           -2.580            -1.950            -1.620
    6           -0.660           -2.580            -1.950            -1.620
    5           -0.712           -2.580            -1.950            -1.620
    4           -0.943           -2.580            -1.950            -1.620
    3           -0.924           -2.580            -1.950            -1.620
    2           -0.961           -2.580            -1.950            -1.620
    1           -0.865           -2.580            -1.950            -1.620

Opt Lag (Ng-Perron seq t) = 18 with RMSE  16.24036
Min SC   =   5.68234 at lag  1 with RMSE  16.93258
Min MAIC =  5.641973 at lag  5 with RMSE   16.6177
```

Appendix 16: KPSS Test for Total Commodities Price (With Trend)

```
Automatic bandwidth selection (maxlag) = 16
Autocovariances weighted by Bartlett kernel

Critical values for H0: tcom is trend stationary

10%: 0.119  5% : 0.146  2.5%: 0.176  1% : 0.216

Lag order    Test statistic
   16           .255
```

Appendix 17: KPSS Test for Total Commodities (without Trend)

```
KPSS test for tcom

Automatic bandwidth selection (maxlag) = 17
Autocovariances weighted by Bartlett kernel

Critical values for H0: tcom is level stationary

10%: 0.347  5% : 0.463  2.5%: 0.574  1% : 0.739

Lag order    Test statistic
   17           2.17
```

Appendix 18: Lag order selection for Cointegration

```
. varsoc e

   Selection-order criteria
   Sample:  1972m5 - 2017m4                      Number of obs      =       540
```

lag	LL	LR	df	p	FPE	AIC	HQIC	SBIC
0	390.451				.013839	-1.44241	-1.4393	-1.43446
1	1311.03	1841.2	1	0.000	.000459	-4.84826	-4.84205	-4.83237
2	1330.66	39.261*	1	0.000	.000429*	-4.91727*	-4.90794*	-4.89342*
3	1331.07	.82448	1	0.364	.000429	-4.91509	-4.90266	-4.8833
4	1331.07	.00035	1	0.985	.000431	-4.91139	-4.89584	-4.87165

```
   Endogenous:  e
    Exogenous:  _cons
```

Appendix 19: Augmented Dickey-Fuller test of Unit root for Cointegration

```
. dfuller e, lag (2)regress

Augmented Dickey-Fuller test for unit root         Number of obs   =       541
```

	Test Statistic	Interpolated Dickey-Fuller 1% Critical Value	5% Critical Value	10% Critical Value
Z(t)	-2.896	-3.430	-2.860	-2.570

MacKinnon approximate p-value for Z(t) = 0.0458

D.e	Coef.	Std. Err.	t	P>|t|	[95% Conf.	Interval]
e						
L1.	-.0219889	.0075925	-2.90	0.004	-.0369035	-.0070743
LD.	.2742824	.0428714	6.40	0.000	.1900662	.3584985
L2D.	-.0391323	.0430854	-0.91	0.364	-.1237689	.0455043
_cons	.0002015	.000887	0.23	0.820	-.001541	.0019439

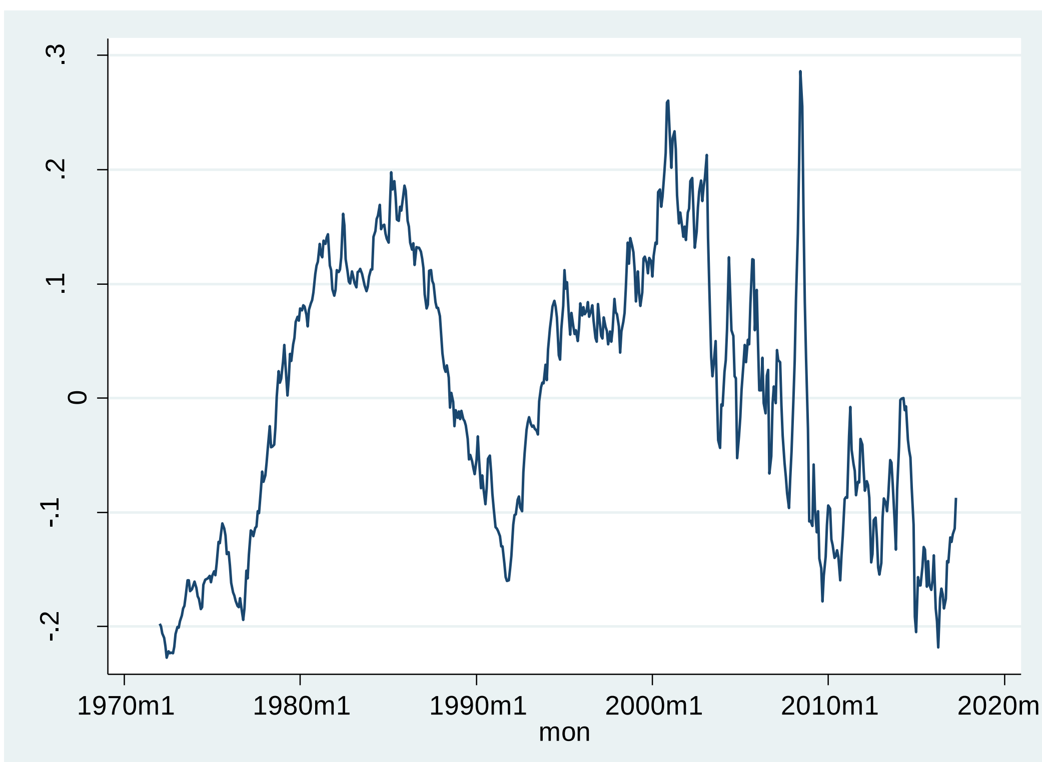
Appendix 20: Regression of Canada-US exchange rate on Total commodity prices

. regress ner mon tcom

Source	SS	df	MS			
				Number of obs	=	544
				F(2, 541)	=	265.56
Model	7.47289203	2	3.73644601	Prob > F	=	0.0000
Residual	7.61188666	541	.014070031	R-squared	=	0.4954
				Adj R-squared	=	0.4935
Total	15.0847787	543	.02778044	Root MSE	=	.11862

ner	Coef.	Std. Err.	t	P>|t|	[95% Conf.	Interval]
mon	.0011277	.000053	21.26	0.000	.0010235	.0012319
tcom	-.0012362	.0000555	-22.27	0.000	-.0013452	-.0011271
_cons	1.16479	.0144223	80.76	0.000	1.136459	1.19312

on



References

Bashar, O. K. M. R., & Kabir, S. H. (2013). Relationship between commodity prices and exchange rate in light of global financial crisis: evidence from Australia. *International Journal of Trade, Economics and Finance, 4(5),* 265-69.

Chen, Y.C., Rogoff, K., and Rossi, B. (2008). Can exchange rates forecast commodity prices? *Technical report, National Bureau of Economic Research.*

Choudhri, E. U., & Schembri, L. L. (2014). Productivity, commodity prices and the real exchange rate; The Long run behavior of the Canada-US exchange rate. *International Review of Economics & Finance, 29,* 537-551.

Ferraro, D., Rogoff, K., & Rossi, B. (2015). Can oil prices forecast exchange rates? An empirical analysis of the relationship between commodity prices and the exchange rates. *Journal of International Money and Finance, 54,* 116-141.

Isard, P. (1977) How far can we push the "Law of One Price"? *The American Economic Review,* 67(5), 942-948.

Jung Y. C., Das, A., & McFarlane, A. (2020). The asymmetric relationship between the oil price and the US-Canada exchange rate. *The Quarterly Review of Economics and Finance, 76,* 198-206.

Kohlscheen, E., Avalos, F. H., & Schrimpf, A. (2017). When the walk is not random: commodity prices and exchange rates. *International Journal of Central Banking, 13(2),* 121-158.

Laidler, D. & Aba, S. (2001). The Canadian Dollar: still a commodity currency. *Backgounder-CD Howe Institute,* 1.

Milijkovic, D. (1999). The law of one price in international trade: A critical review. *Applied Economic Perspectives and Policy, 21*(1), 126-139.

Mow, M. (2013). *The Canadian dollar as a commodity currency: Has the relationship changed?* (Doctoral dissertation, University of Victoria).

Officer, L. H. (1986). The law of one price cannot be rejected: two tests based on the tradable/nontradable price ratio. *Journal of Macroeconomics, 8(2),* 159-182.

Richardson, J. D. (1978). Some empirical evidence on commodity arbitrage and the law of one price. *Journal of International Economics, 8(2),* 341-351.

Rogoff, K. (1996). The purchasing power parity puzzle. *Journal of Economic Literature, 34(2),* 647-668.